



Figure 2: Traffic light LEDs module

driving the different colours of external LEDs, ensuring the red, amber, and green sequence is on, respectively, with a delay in between.

--- Start of trafficlight.py ---

```
from machine import Pin, reset
from time import sleep

tfl1 = {}

tfl1['rled'] = Pin(0, Pin.OUT) # R, GPIO 0, Pico Pin 1
tfl1['gled'] = Pin(1, Pin.OUT) # G, GPIO 1, Pico Pin 2
tfl1['yled'] = Pin(2, Pin.OUT) # Y, GPIO 2, Pico Pin 4
                                     # G, GND,
Pico Pin 3

def rstlds(tfl):
    tfl['rled'].off()
    tfl['gled'].off()
    tfl['yled'].off()

def stoponred(tfl):
    tfl['rled'].on()
    tfl['gled'].off()
    tfl['yled'].off()

def gongrn(tfl):
```

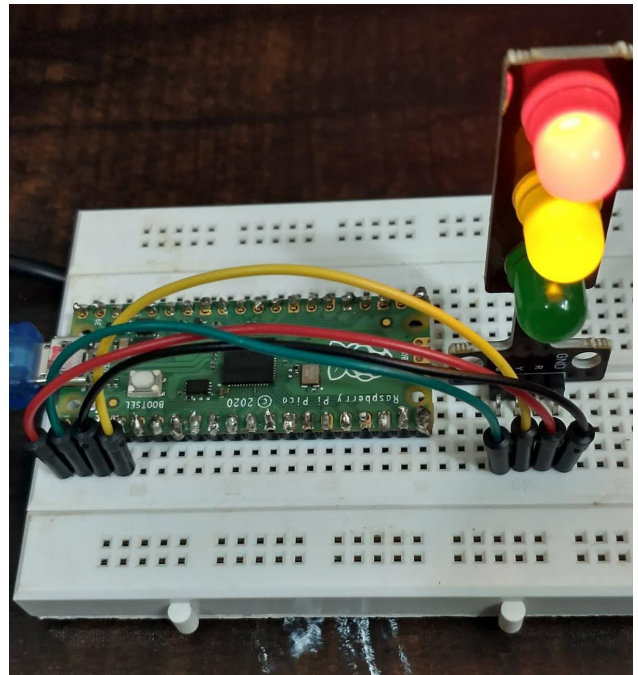


Figure 3: Wired circuit of external LEDs traffic light

```
tfl['rled'].off()
tfl['gled'].on()
tfl['yled'].off()
```

```
def readyonlwl(tfl):
    tfl['rled'].off()
    tfl['gled'].off()
    tfl['yled'].on()
```

```
def readybgrn(tfl):
    tfl['gled'].off()
    tfl['yled'].on()
```

```
while True:
    try:
        stoponred(tfl1)
        sleep(5)
        readybgrn(tfl1)
        sleep(2)
        gongrn(tfl1)
        sleep(5)
        readybgrn(tfl1)
        sleep(2)
    except KeyboardInterrupt as e:
        rstlds(tfl1)
        reset()
```

--- End of trafficlight.py ---